

Deep builder

**skip unused Lanczos; $k = 8$ live;
 $k = 10/11/12$ are not ABD-living**

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Abstract

r443 could not form $\delta_k = 1 - q^\dagger$ at selected $k = 10$ ($k_z 197$, $N = 4071$): the published caller timed out at 20 s inside `HS.window.data`'s unused Lanczos, before `border_chain_pack` ran. This note profiles that hotspot, replaces it with a bit-identical atom construction, rebuilds $k = 8$ live (closing the r427 circularity), and attempts $k = 10, 11, 12$. The three deep windows are *not* ABD-living: $k = 10, 11$ complete the $\tilde{\mu}$ -chain with a terminal sign-flip, $k = 12$ aborts. They do not extend the r443 sequence. The ABD-ok slice $k = 5..9$ with live $k = 8$ still prefers a floor ($\delta_\infty = +0.02741$, $\Delta\text{AIC} = 5.11$). No RH claim.

Status. No new SATZ (the $q^\dagger$ formula is r442). Propositions 1–4 are a named infra census. Ausgang `INFRA_UNLOCKED` / `K8_CROSS_CONFIRMED` / `DEEP_NOT_ABD_LIVING` / `SLICE_FLOOR_PREFERRED` / `FULL_SEQUENCE_UNSETTLED` / `COFINAL_OPEN`. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 Hotspot

Proposition 1 (Unused Lanczos). *The r443 20 s timeout at $k = 10$ is `HS.window.data`'s two-sided Lanczos ($O(N^2M)$ reorthogonalization), which is not an input of $q^\dagger = B_w^{-1}b^T(I - C)^{-1}b$. The folded $\pm$ atoms of the sealed smooth comb are bitwise identical with or without that chain.*

Measured on this machine (2026-08-30):

	$k = 9$ ($N = 1433$)	$k = 10$ ($N = 4071$)
<code>HS.window.data</code> (Lanczos)	0.768 s	20.57 s
slim atoms (folded $\pm$ only)	0.081 s	0.18 s
<code>BH.bord_chain</code>	0.062 s	0.29 s
numpy-slim pack (with den)	0.53 s	4.4 s

Speedup at $k = 10$: the old path never reached C ; the new path returns $q^\dagger$ in 0.8 s (den add-on 3.6 s). Canonical engine is numpy (same `np.sum` order as ABD/V). numba reproduces the three-term algebra with summation-reorder drift $|dq| = 5.6 \cdot 10^{-9}$ at $k = 9$ (bar 10^{-8}); it is *not* faster on the chain (1.45 s vs 0.062 s at $k = 9$) and is reserved for matrix-free CG.

Bit-gate $k = 3..9$: $|dq| = |dd| = 0$ at 10^{-12} vs the published ABD/V path; den/ Σ likewise; r443 six-digit δ pins reproduced. mpmath dps-40 at $k_z 17$: $|q_{64} - q_{\text{mp}}| = 1.1 \cdot 10^{-16}$ (we solve $(I - E)x = v$, no `eigvalsh`; r427 ward). Determinism: `run1 = run2` at $k_z 17$.

2 $k = 8$ cross

Proposition 2 ($k = 8$ live). *Selected $k = 8$ ($k_z 69$, $N = 5690$) rebuilds in 8.2s and matches the r421 pin: $R = 0.047628$, $\text{den} = 1.568548$, $\Sigma = 0.383825$, $P1$. Hence $\delta = 0.047488$, $q^\dagger = 0.952512$. The r427 circularity (“ $k = 8$ not rebuilt”) is closed.*

The chain stays positive ($\min \eta = 2.6 \cdot 10^{-14}$). $|Y| = 3089$, dense $Y \times Y$ residual $2.6 \cdot 10^{-15}$.

3 Deep windows

ABD.ok requires the $\tilde{\mu}$ -chain to reach depth N with every $h_k > 0$. That gate, not runtime, is what blocks $k = 10+$.

Proposition 3 (Not ABD-living). *On the selected sequence: $k = 10$ ($N = 4071$) completes with 107 sign-flips from $n = 3788$; $k = 11$ ($N = 12508$) completes with 839 sign-flips from $n = 10581$, and signed B_w explodes to 21434; $k = 12$ ($N = 45444$) aborts at $n = 12737$ (η underflow). None of the three is the r443 object.*

Formula-literal $q^\dagger = B_w^{-1} b^T (I - C)^{-1} b$ with the completed signed chain (not used in the AIC):

k	N	n_{flip}	δ (signed B_w)	δ_{pos}	branch
10	4071	107	0.000419	0.031823	VAC
11	12508	839	0.996548	−2.29	VAC
12	45444	abort	—	—	—

$k = 10$ $\text{den} = 1.558$, $R = 0.00838$, residual $9.5 \cdot 10^{-12}$. $k = 11$ $\text{den} \approx 1.0007$, $R \approx 0.997$ tracks the exploded budget, not a floor datum. $k = 12$ has no faithful B_w at full depth (B would be 8.5 GB).

Dead χ_{3-15} with the new builder: $\delta = -0.023181 = \text{old (control)}$.

4 Fit, both abscissae

Three diagnostic models (r421, not a proof): M_1 floor $+ck^{-a}$, M_2 pure power, M_3 $c/\log$. Primary sequence = ABD-ok selected windows (the r443 object), now including live $k = 8$.

Proposition 4 (Slice with live $k = 8$). *On $k = 5, 6, 7, 8, 9$, M_1 wins for δ with $\delta_\infty = +0.02741$ (ΔAIC vs M_2 is 5.11). Without $k = 8$ this was $+0.02670 / 6.01$. Live $k = 8$ sits on the curve and does not flip the slice verdict. On the full ABD-ok $k = 3..9$, M_3 still wins (the $k = 5$ bump 0.109). Against N , M_2 wins: N is not monotone (r427, illegal abscissa). Drop- $k = 5$ flips to M_2 ; drop- $k = 9$ keeps M_1 but jumps δ_∞ to 0.039.*

$k = 10/11/12$ do not extend the ABD-ok sequence, so they do not decide floor vs decay. A formula-literal $k = 10$ point at $\delta = 4 \cdot 10^{-4}$ would drive M_1 onto the grid floor -0.02 ; that is a sign-flip artefact, not a Selected datum.

The liminf itself stays open. If a floor $\delta_\infty > 0$ exists then eventually $q^\dagger \leq 1 - \delta_\infty$ and $\kappa^\dagger = 0$ by r442 — tautological, not a proof.

Claim boundary

Nothing here is evidence for or against the Riemann hypothesis. The identities are finite linear algebra. The deep census is a named construction-gate, not a theorem of RH.

References

- [1] Bordered chain / $B_w = S_{N-2} + 5/7$.
- [2] R -floor AIC; $k = 8$ pin.
- [3] Campaign audit: N is not monotone; $k = 8$ not live (closed here).
- [4] $q^\dagger = B_w^{-1}b^T(I - C)^{-1}b$.
- [5] Selected δ -floor; $k = 10$ timeout.